

On the Design of a Social Norm Intervention to Promote Interethnic Cooperation through Tolerance: An Agent-Based Model

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Overview

- Background & criticisms
- Our research objective
- A micro-theory
- An agent-based model
- Some doubts

Background & criticisms

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- However, these interventions almost exclusively aim to achieve these outcomes by reducing individual-level prejudice (Paluck et al., 2021).
- Why would that not be sufficient?

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- Group-based prejudice, as the differential treatment of others because they belong to an ethnic out-group, is, of course, an important one.
- Another increasingly outlined motivation is people's principled disapproval of outgroup beliefs, values, and/or practices, which can worsen interethnic relations independently of prejudice (Verkuyten et al., 2023).

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- That is, those who are not prejudiced can still be intolerant and act accordingly (Verkuyten et al., 2023).
- Although this motivation is perhaps more morally justifiable than a prejudicial one, it can still lead to worsened interethnic cohesion.

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- Instead, interventions should promote value-based reasons to accept these beliefs and practices despite disapproval
- and aim to foster tolerance, as a behavioral intention to not interfere with the disapproved of belief, value or practice (Verkuyten, et al, 2023; Simon, 2023).

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- They must also account for reasonable sources of disapproval and manage their consequences through promoting tolerance (Verkuyten et al., 2023).

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- For example, they aim to decrease individual-level prejudice through cognitive and emotional training.
- But they typically ignore how the social structures in which people are embedded shape whether those changed attitudes persist, or translate into improved behavior.
- Individual-level interventions in the real-world typically show effects that are limited in size and duration compared to the laboratory (Paluck et al., 2021).

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- What's needed, therefore, is a relational and longitudinal inquiry into how real-world, interventions can lead to sustained improvements in interethnic attitudes and behaviors.

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- Put differently, our objective is to identify designs under which a tolerance intervention might be effective in promoting interethnic cohesion in the real-world.
- Interethnic cohesion → increased and sustained interethnic tolerance and cooperation.

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- We use it as a forecasting tool to explore how micro-level mechanisms play out to produce different levels of interethnic cohesion on the macro-level given different intervention designs.
- Its crucial feature is that it models the micro–macro link and helps us study emergent effects in interethnic cohesion that occur after a change in tolerance has been introduced in the network.

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- Students in German high schools.
- We specifically concentrate on increasing tolerance among ethnic majority members (e.g., Native German people) toward the practices, values, and beliefs of ethnic minority members (e.g., Turkish German people).

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- Our micro-theory involves three concepts: tolerance as an attitude, and cooperation and friendship as social network relationships.
- We begin by assuming a hypothetical intervention (unspecified in form) that increases individual-level tolerance by increasing equality-based respect as a counterweight to disapproval among a subset of actors.
- Equality-based respect as "recognizing and treating outgroup individuals as equals, with the same rights and dignity as oneself" (Renger & Reese, 2017).

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- In theory, therefore, tolerance increases when equality-based respect increases relative to disapproval.
- Key assumption: since we model respect tolerance directly, we must assume that some degree of disapproval is already present, otherwise, there would be nothing to tolerate.

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- Social influence refers to the process by which individuals adjust their attitudes as a result of interaction with another individual (Rashotte, 2007).
- We focus specifically on friends (rather than cooperation partners) because the social influence of friends on interethnic attitudes has relatively strong theoretical and empirical grounding (e.g., Simons, Jaspers, & Van Tubergen, 2025).

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- no shift occurs when the difference is not too small or too large
- and more likely to polarize away when the attitude difference is sufficiently large.
- Although empirical evidence for this mechanism is mixed (Tang, Snijders, & Flache, 2025), it is an attractive option because it allows for repulsion on tolerance instead of just assimilative influence.

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- A subset of ethnic majority individuals receives the intervention, leading to an increase in their tolerance.
- These individuals may then influence their friends to also become more tolerant, but only if their new attitudes fall within their friends' latitude of acceptance.
- Alternatively, no change or a decrease in tolerance may occur.

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- Selection refers to how people decide to structure their relationships with others, given their level of tolerance.
- Theoretically, increased tolerance should “only” result in decreased negative outgroup behaviors (e.g., bullying) (Verkuyten et al., 2023), since students now refrain from acting on their disapproval.

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- We instead postulate that increased tolerance can result in increased interethnic cooperation.

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- then if a friend influences you to adopt a more tolerant attitude, you are more likely to cooperate with an ethnic outgroup member due to a wider radius of trust.

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- Influence: Friendship with ethnic ingroup members influences tolerance based on an attraction-repulsion mechanism.
- Selection: The level of tolerance determines the probability of cooperation with outgroup members.

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- And for several endogenous network effects, e.g., reciprocity, transitivity, degree effects, and one exogenous effect: gender choice homophily.

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- Here, we assume that the lack of effectiveness was solely due to the intervention design.
- In other words, the intervention might have worked if a different design had been implemented.

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- But that is not very practical.
- Instead, we use an ABM as an alternative strategy.

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 - Complex contagion → adoption carries social risk and therefore requires multiple simultaneous exposures.
 - Delivery strategy: Should interventions be applied in a clustered or random fashion? → related to contagion type.

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- Manipulate initial conditions for each cell in the intervention matrix to examine outcomes in tolerance and cooperation across the different scenario's.

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- i.e., influence only activates given sufficient simultaneous exposures from friends.

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- Pick one?
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- Or maybe focus on the school level instead?

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- Doubt 3: What should we do if the micro-theory is superficially or not supported?
- Say that we obtain support for the micro-theory in only a few classes after an extensive search for a sufficient fit, would there be a concern of ad hoc fitting?
- Or say we don't find any support at all, should we then formulate and test an alternative micro-theory (for example on negative ties?).

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- What actually happens to ethnic outgroup trust when someone becomes more tolerant?
- Maybe someone is aware of any relevant reading on this topic?

Thank you.